



Society of Petroleum Engineers

SPE-230096-MS

Pioneering UAE-Built 5G-Enabled IoT for ADNOC's Digital Oilfield: A Collaborative Journey

R. Espinoza, GOEasyFlow, UAE; M. Majer, ADNOC Upstream, UAE

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230096-MS](https://doi.org/10.2118/230096-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Objectives/Scope: The objective of this work is to present the innovative journey behind the development of a 5G-enabled wireless monitoring and control technology, fully designed and manufactured in the UAE, to meet the operational and cybersecurity requirements of ADNOC's upstream assets. The project addresses the challenge of enabling real-time operational awareness and cost-effective data acquisition in remote and industrial environments, leveraging ADNOC's 5G network as the foundational connectivity layer.

Methods, Procedures, Process: The development followed a structured process from concept to lab validation, including joint definition of technical requirements, iterative prototyping, integration of dual band 5G modules, and alignment with ADNOC's cybersecurity protocols. The solution employs low-band 5G for ultra-low-power, battery-operated IoT sensors, and high-band 5G for high-frequency, high-throughput gateways. Validation was conducted at ADNOC's 5G Lab. Key phases included hardware development, network integration, cybersecurity vetting, and controlled lab testing, alongside the design of a full-field deployment architecture.

Results, Observations, Conclusions: The system demonstrated full compliance with ADNOC's industrial and cybersecurity standards and achieved reliable performance under rigorous lab testing. The dual band 5G configuration proved technically sound and operationally well-suited for upstream oilfield applications, supporting both real-time monitoring and remote-control functions. The final solution architecture offers significant potential to enhance operational awareness, streamline infrastructure, and reduce the cost and complexity of data acquisition across ADNOC's fields.

Novel/Additive Information: This work highlights a first-of-its-kind collaboration between GOEasyFlow and ADNOC to deliver a UAE-built, 5G-native industrial IoT technology tailored for oilfield automation. It demonstrates how operator-driven innovation, developed in-country and aligned with local infrastructure and cybersecurity mandates, can accelerate digital transformation. The dual-band approach and local manufacturing model present a replicable blueprint for other oil companies seeking secure, scalable, and cost-effective wireless monitoring and control solutions.

Introduction

The increasing complexity of ADNOC's upstream operations has amplified the demand for reliable, secure, and low-power field instrumentation. Existing wired solutions, while secure, entail high CAPEX and significant commissioning times often extending to months or years. Wireless LPWAN technologies like LoRaWAN have addressed range and power requirements. However, concerns around open-spectrum cybersecurity have remained a barrier for mission-critical applications. This project addresses that gap by integrating LTE-M and LoRaWAN-over-5G into ADNOC's secured 5G infrastructure, enabling the operational efficiency of both wireless systems with the security standards expected of critical OT networks.

ADNOC's increasing reliance on IoT-based field instrumentation for operational efficiency and environmental compliance has brought forward the need for secure, reliable, and low-power wireless communication technologies. While LoRaWAN has demonstrated excellent range and power efficiency in remote oilfield deployments, concerns remain about standalone cybersecurity posture and potential data interception in open-spectrum networks.

To address these challenges, ADNOC's Digital Unit, in collaboration with GOEasyFlow, developed and tested a first-of-its-kind, UAE-designed and manufactured 5G-enabled IoT monitoring and control solution. This system integrates LTE-M (Cat-M1) and LoRaWAN-over-5G within ADNOC's secured 5G infrastructure, combining local R&D, firmware, and hardware design to produce a solution tailored for upstream oil and gas operations.

The solution leverages low-frequency LTE-M Band (700 MHz), offering extended range, better penetration, and low power consumption suitable for long-term battery-powered deployments. The Phase 1 test was conducted at ADNOC's 5G Lab, focusing on LTE-M performance with real-time Modbus polling, MQTT data publishing, relay control, and edge computing logic.

In-Country Value (ICV) Impact

This project directly contributes to ADNOC's In-Country Value (ICV) program by ensuring the entire design, prototyping, and manufacturing process was executed in the UAE. All hardware was engineered locally, firmware developed by in-country talent, and assembly performed at UAE-based facilities. This approach not only accelerates development cycles by shortening supply chains but also builds sustainable local capability in advanced industrial IoT manufacturing. The resulting solution represents a replicable model for fostering high-technology industries within the UAE while delivering operational benefits to ADNOC's upstream assets.

Methodology

This project is structured into multiple phases, with Phase 1 covering all the work completed to date and documented in this paper. Future phases will progressively advance the technology toward full commercial field deployment:

Phase 2: Design, manufacture, and testing of industrial-grade 5G IoT battery-powered instruments for non-ATEX environments.

Phase 3: ATEX certification for hazardous-area deployment.

Phase 4: Full-scale rollout across ADNOC's operational assets.

Phase 1 was executed in two major test parts, both performed under controlled conditions at ADNOC's 5G Lab in Abu Dhabi, following the success criteria and procedures defined in the official Phase 1 Test Protocol.

Part 1 – LoRaWAN-over-5G VxLAN Test

The first stage of Phase 1 assessed the feasibility of operating existing LoRaWAN instrumentation within ADNOC's 5G infrastructure by encapsulating its traffic inside a Virtual eXtensible LAN (VxLAN) tunnel.

This approach allowed unmodified LoRaWAN field devices to benefit from 5G-grade cybersecurity without changing their firmware or hardware.

As shown in Figure 1 [Network Topology Diagram], LoRaWAN sensor data originated from field-connected devices and passed through a LoRaWAN controller linked to a Nokia FastMile mm29-b customer premises equipment (CPE). The CPE established a secure VxLAN tunnel to the 5G core, where traffic flowed through multiple defense-in-depth layers, including GTP transport, IPSec encryption, OT firewall segmentation, and VLAN isolation, before reaching ADNOC's operational monitoring systems. This architecture ensured that every LoRaWAN packet was carried securely over the private 5G network, eliminating risks typically associated with open-spectrum LPWAN deployments.

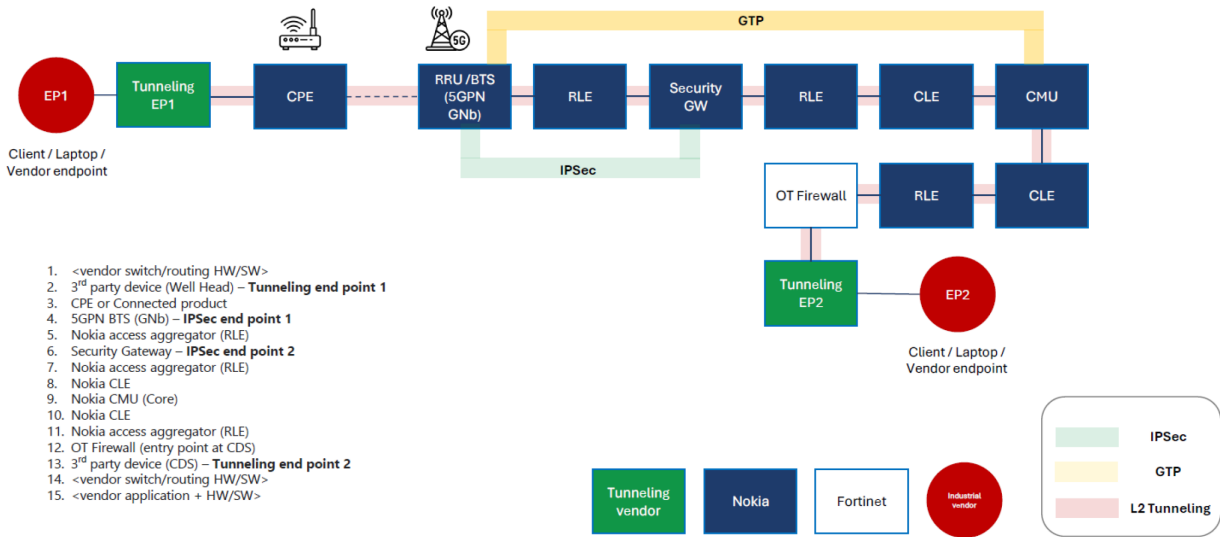


Figure 1—Network Topology Diagram

The physical integration is illustrated in Figure 2 [Lab Test Setup Photos], where the CPE and LoRaWAN controller were connected to various industrial endpoints:

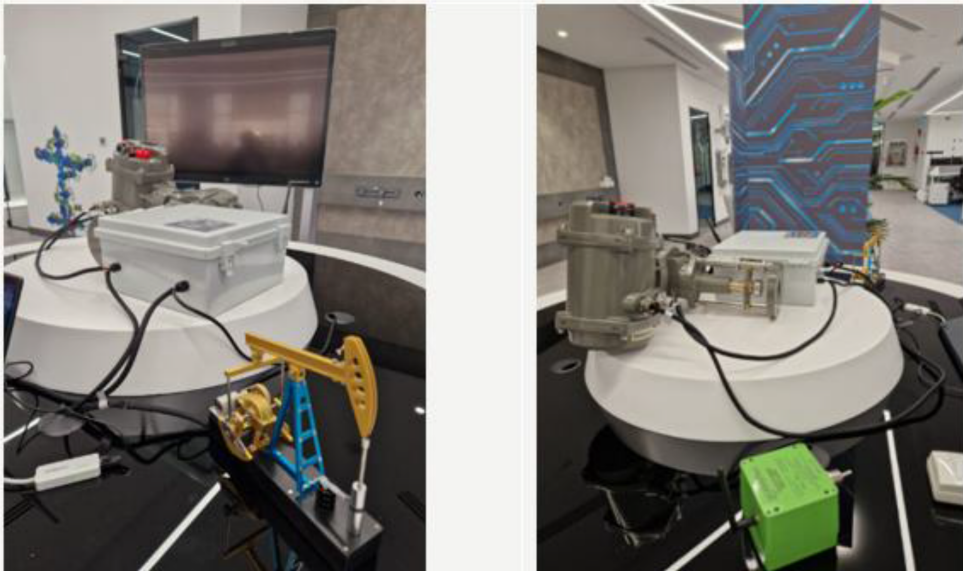


Figure 2—Lab Test Setup Photos – Wired to LoRaWAN controller, Temperature Transmitter, Actuator, Ambient Monitor, and Sucker Rod Pump Model

- A temperature transmitter for continuous environmental monitoring.
- An actuator for control function testing.
- An ambient condition monitor for local safety and environmental readings.
- A scaled sucker rod pump model to simulate upstream oilfield mechanical operations.

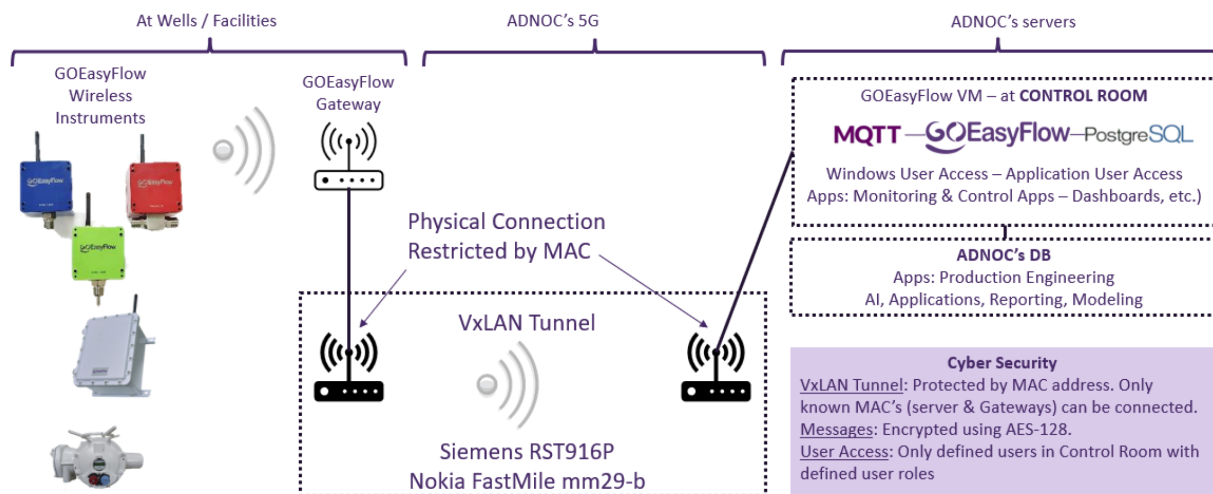


Figure 3—LoRaWAN-over-5G VxLAN Integration

From a cybersecurity perspective, the LoRaWAN-over-5G test environment shared the same ADNOC 5G core network as the LTE-M tests, but the VxLAN tunnel was an additional security feature unique to this scenario. This tunnel encapsulated LoRaWAN packets end-to-end from the CPE to the secure network termination point, ensuring full traffic integrity and confidentiality.

Outcome

The VxLAN implementation proved fully transparent to LoRaWAN MAC-layer functions such as join requests, uplinks, and downlinks. No packet loss, latency spikes, or functional disruptions were observed that could be attributed to the encapsulation process. This result demonstrates that existing LoRaWAN deployments can be migrated into ADNOC's secured 5G environment without replacing hardware, providing an immediate pathway to improved cybersecurity while maintaining operational continuity.

Part 2 – LTE-M (Cat-M1) Test on Low Band

The second part of Phase 1 focused on validating a newly developed LTE-M-enabled operational technology (OT) monitoring device, fully designed and manufactured in the UAE. This device integrates Modbus RTU polling, MQTT communications, relay actuation, and edge computing logic into a single, battery-powered, field-ready unit. The test aimed to confirm that LTE-M, operating within ADNOC's 5G core on Low Band (700 MHz), could meet the stability, latency, and cybersecurity requirements for upstream oilfield monitoring and control.

Objective

To assess the LTE-M link's stability, latency, and compatibility with ADNOC's OT infrastructure for real-time industrial monitoring and remote control, while verifying low-power operation suitable for battery deployment.

System Architecture and Communication Flow

The LTE-M system architecture is shown in Figure 4, illustrating how the GOEasyFlow development board connected to industrial sensors and actuators communicates through the ADNOC 5G core. The device registered to the LTE-M network (Low Band), passed through Nokia's access and core elements (RLE, CLE, CMU), and traversed Fortinet security gateways before reaching the OT firewall and MQTT broker. All communications remained within ADNOC's secure core, ensuring compliance with D&CS cybersecurity standards.

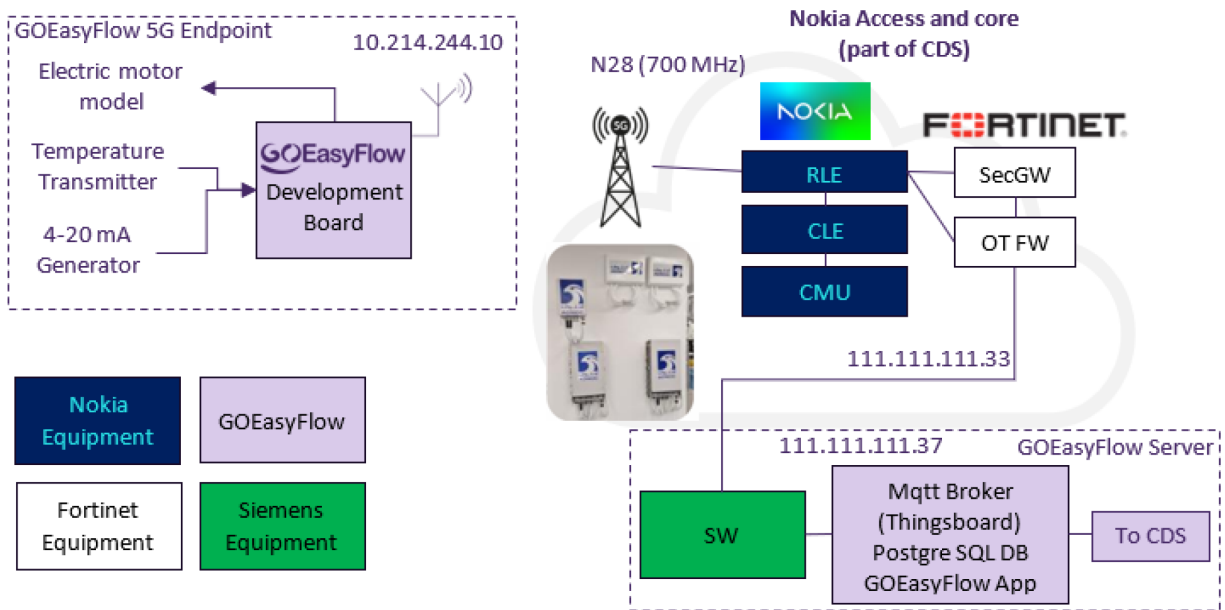


Figure 4—LTE-M System Architecture

Firmware Architecture

The device's firmware stack (Figure 5) was organized into four layers:

- Application Layer – Modbus polling handler, edge control logic, relay control (manual override via MQTT), AES encryption, and JSON data assembly.
- Communication Layer – LTE-M modem control via TinyGSM, AT command handling for registration, MQTT subscription management.
- Hardware Abstraction Layer – RS485 Modbus driver, GPIO relay control, power management routines.
- Hardware Platform – MCU, LTE-M (Cat-M1) modem, external antenna, and SIM card.

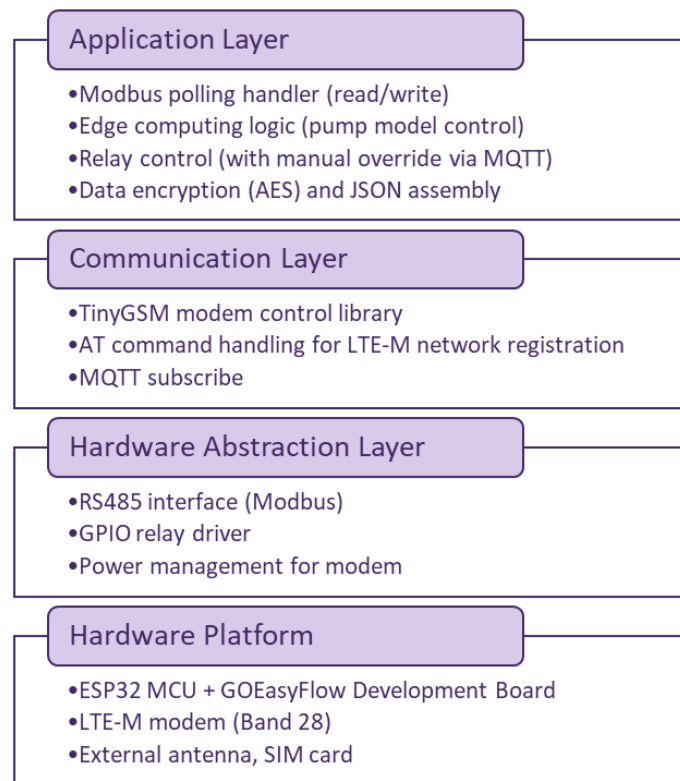


Figure 5—LTE-M Firmware Architecture

Hardware Integration

The hardware integration layout (Figure 6) highlights the modular design:

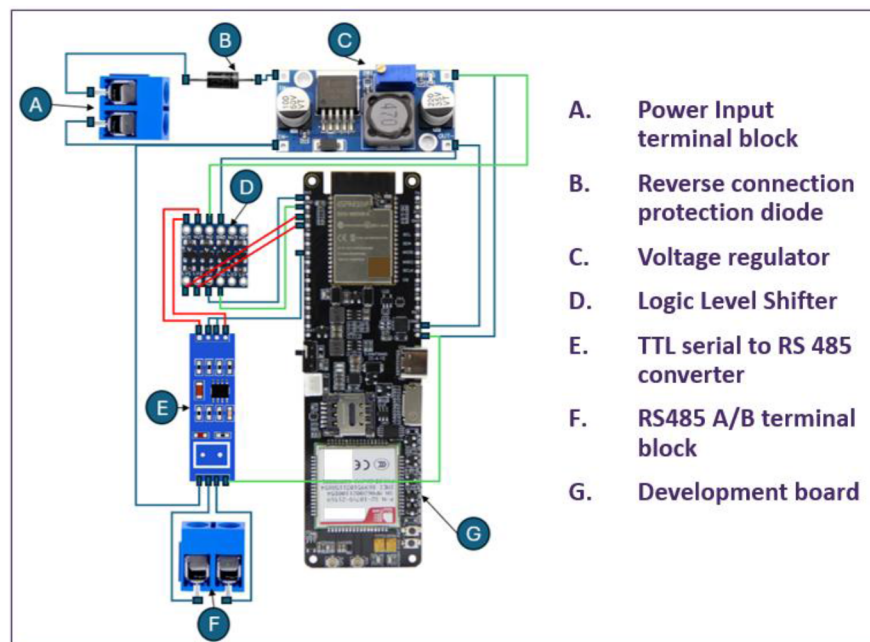


Figure 6—Device PCB Photo

- A–C: Power input terminal, reverse polarity protection, and voltage regulation.
- D–E: Logic level shifting and TTL-to-RS485 conversion for Modbus compatibility.

- F–G: RS485 terminal block and main development board with LTE-M modem.

The fully assembled PCB is shown in [Figure 7](#), enclosed in a ruggedized, IP-rated housing designed for oilfield use.

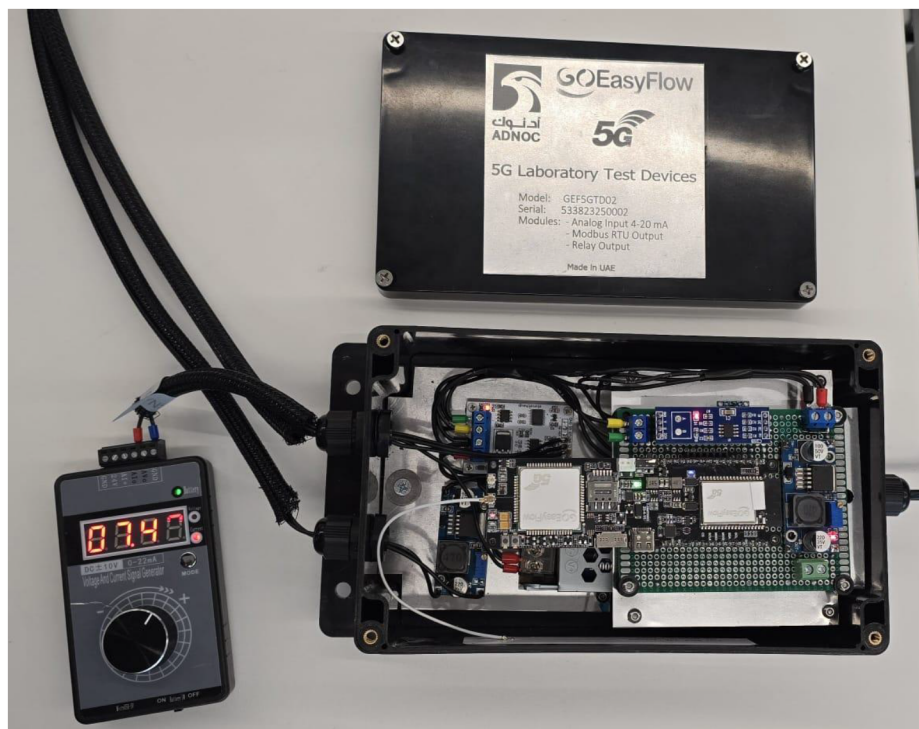


Figure 7—5G LTE-M Test Device

Objective: To evaluate the LTE-M link's stability, latency, and compatibility with ADNOC's OT systems for real-time monitoring and control.

Hardware

- LTE-M radio modem operating on Low Band (700 MHz) for superior coverage and lower power draw.
- Microcontroller running dual-radio firmware to support LTE-M and optional LoRaWAN modes.
- RS485 transceiver for Modbus polling of industrial sensors.
- Relay driver circuit for local or remote actuation.
- Power management module designed for long-term battery operation.

Test Procedures

LTE-M Registration > MQTT Connectivity > Modbus Polling > Relay Control > Edge Computing

The LTE-M test followed a sequential procedure replicating real-world operational workflows:

1. LTE-M Registration – The device powered on and attached to ADNOC's LTE-M Low Band network, confirming immediate registration without retries.
2. MQTT Connectivity – A secure session was established with the broker inside ADNOC's 5G core, enabling continuous publish/subscribe operations.

3. Modbus Polling – Continuous RS485 polling was performed at varied intervals to acquire readings from simulated field sensors, including temperature transmitters and current loop generators.
4. Relay Control – Remote actuation commands were sent via MQTT downlink, with instant execution and acknowledgment observed.
5. Edge Computing – Local threshold-based logic controlled the pump model autonomously, ensuring continued operation even if network connectivity was lost.

Instrumentation Setup: As shown in Figure 7, the test bed incorporated the LTE-M modem, microcontroller, and industrial I/O into a compact enclosure connected to:

- An electric motor model simulating mechanical load.
- A temperature transmitter.
- A 4–20 mA signal generator representing process instrumentation.

Outcome: During the high-frequency phase of the LTE-M tests, stable posting intervals of 1.0–1.1 seconds were achieved without packet loss. All success criteria (registration, connectivity, polling, control, edge logic) were met, confirming readiness for industrial deployment. This performance also supports low-duty-cycle modes for extended battery life, a key requirement for remote upstream installations.

Cybersecurity Compliance

Both the LoRaWAN-over-5G and LTE-M test configurations operated within the same ADNOC 5G core network, ensuring that security policies, monitoring, and traffic management applied uniformly across both scenarios. This common network environment is governed by ADNOC's Group Digital & Cybersecurity (D&CS) standards, which mandate multi-layer defenses and strict segmentation of operational technology (OT) traffic.

Security measures implemented include:

- AES-128 end-to-end payload encryption (RAW_PAYLOAD) with decryption only at the application server (DECODED_PAYLOAD).
- VLAN segregation to restrict device access to authorized OT networks.
- Multiple firewall layers in ADNOC's D&CS-compliant architecture.
- Intrusion detection and prevention systems within the 5G core.

For the LTE-M configuration, device traffic was encapsulated within GTP tunnels over the 5G radio link, with IPSec encryption providing additional confidentiality and integrity protection. VLAN segmentation and firewall rules were enforced end-to-end, isolating the device's data path from all non-authorized network zones.

For the LoRaWAN-over-5G test, an additional VxLAN tunnel was implemented between the Nokia FastMile mm29-b CPE and the 5G core. This encapsulated the LoRaWAN gateway traffic, allowing legacy LoRaWAN devices to operate securely without modification, while inheriting the same firewall segmentation, VLAN isolation, and intrusion detection measures as LTE-M traffic.

Beyond transport-level protections, all raw sensor payloads—regardless of radio technology—were encrypted end-to-end with AES-128 prior to transmission. Physical and logical access to network termination points remained restricted through ADNOC OT VLAN controls.

This architecture not only satisfies ADNOC's industrial cybersecurity requirements but also eliminates the vulnerabilities typically associated with standalone LPWAN deployments, providing a defense-in-depth posture capable of supporting critical oilfield operations in both new and legacy device scenarios.

Results and Data Interpretation

Part 1 – LoRaWAN-over-5G VxLAN Test

The first part of Phase 1 validated the seamless operation of existing LoRaWAN-based field devices within ADNOC's secured 5G core network, without requiring any firmware or hardware modifications.

RSSI and SNR readings remained stable, and MAC-layer operations were unaffected. This demonstrates that legacy LPWAN assets can gain 5G-grade security with zero hardware or firmware changes.

As illustrated in Figure 3, the setup used a GOEasyFlow LoRaWAN gateway connected to the Nokia FastMile mm29-b customer premises equipment (CPE), which established a VxLAN tunnel to ADNOC's 5G core. This tunnel, protected by MAC address restrictions, created a point-to-point encrypted path from the gateway to the control room, ensuring that only authorized gateways could connect.

Once inside ADNOC's network, LoRaWAN traffic traversed the 5G core infrastructure with the same firewall segmentation, VLAN isolation, and intrusion detection applied to other OT traffic. Importantly, the 5G core network used in this test was the same as that employed for LTE-M, meaning both technologies benefited from the same ADNOC cybersecurity framework. For LoRaWAN, the VxLAN tunnel was an added security layer, compensating for LPWAN's inherent open-spectrum vulnerabilities.

The transmitted data maintained end-to-end AES-128 encryption at the application layer. The LoRaWAN payload (RAW_PAYLOAD) arrived at the server still encrypted and was only decrypted server-side to produce the DECODED_PAYLOAD, as shown in Table 1. This confirmed that data confidentiality was preserved throughout the transmission path, even when traversing multiple network segments.

Table 1—MQTT LoRaWAN Payload Example

TIMESTAMP	RAW_PAYLOAD	DECODED_PAYLOAD	deviceName	gatewayID	rssi	loRaSNR	Txinfo frequency
6/23/2025 13:14:10	YldfgEEBAITbUF	62575E7E01040402004DB505	PCB_001	0102e4956e4e8f01	-40	10	867700000
6/23/2025 13:14:44	TG9SYSBqb2luZWQgdXBsaW5rAA==	4C6F5261206A6F696E65642075706C696E6800	PCB_001	0102e4956e4e8f01	-38	8	867500000

Performance Observations

Throughout the test, LoRaWAN MAC-layer operations including join requests, uplinks, and downlinks, remained fully functional. No packet loss or abnormal latency was observed that could be attributed to VxLAN encapsulation. The tunnel proved transparent to device operation, meaning that sensor behavior, timing, and protocol integrity were unaffected.

Practical Implications

This outcome confirms that ADNOC can migrate existing LoRaWAN deployments into its 5G network without replacing field devices or altering their configuration. By leveraging VxLAN tunneling, legacy devices instantly gain 5G-grade network protections while maintaining their operational profile. This approach offers a cost-effective, low-disruption path toward unified 5G-secured field communications.

Key Findings – LoRaWAN-over-5G VxLAN Test

- No degradation in performance when encapsulated over VxLAN; provides immediate cybersecurity benefits for legacy devices.
- No observable packet loss or latency increase introduced by the VxLAN encapsulation.
- End-to-end AES-128 encryption confirmed: RAW_PAYLOAD arrives encrypted, DECODED_PAYLOAD generated only at the server.

- All LoRaWAN MAC-layer operations (join, uplink, downlink) remained fully functional without firmware or hardware changes.
- Tunnel implementation was transparent to field devices, requiring no configuration changes on the sensors or gateways.
- Transparent migration path identified for legacy LoRaWAN deployments to gain 5G-grade cybersecurity without replacing existing equipment.

Part 2 – LTE-M (Cat-M1) Test on Low Band

The second part of Phase 1 focused on the LTE-M-enabled OT monitoring device, integrating Modbus RTU, MQTT, relay outputs, and edge computing. Testing covered multiple success criteria to replicate real-world oilfield applications.

Edge computing was validated using a tank-level control simulation: when local thresholds were reached, the device's relay output activated pump controls without external server commands. This demonstrates autonomous decision-making at the edge, which can extend to use cases such as gas lift valve modulation, chemical injection optimization, and solar panel tracking.

The high-frequency phase (Segments 16–25 in the data log) showed posting intervals consistently between 1.0 and 1.1 seconds, proving that the LTE-M link can sustain rapid, reliable updates while remaining compatible with low-duty-cycle battery-powered designs. Testing also revealed the relationship between firmware timing parameters and posting reliability:

- Stable operation required a minimum delay of 50 ms for both the LTE-M radio module and Modbus polling.
- Instability was observed when either delay dropped below 50 ms, particularly in combinations of 10 ms LTE-M radio module delay and <50 ms Modbus polling.
- JSON payload assembly time needed to be ≥ 100 ms to prevent data posting interruptions.

Table 2—LTE-M Lab Test Results (Final High-Frequency)

Success Criteria	Description	Pass/Fail	Final Phase Observation
LTE-M Registration	Device must successfully attach to ADNOC LTE-M low Band network	☑ Pass	Instant registration with MCC/MNC confirmed; no drops in final phase
MQTT Connectivity	Establish and maintain MQTT broker link	☑ Pass	Stable MQTT session maintained; posting interval stabilized at ~1.0–1.1 s
Modbus Polling	Continuous RS485 polling of sensor data	☑ Pass	Continuous polling without data gaps; all readings captured in sequence
Relay Control (Remote)	Remote control via MQTT downlink	☑ Pass	Commands executed successfully and acknowledged instantly in final phase
Edge Computing Logic	Local pump-control logic with threshold setpoints	☑ Pass	On/off control at preset thresholds functioned correctly; no missed triggers

Key Findings

- LTE-M: Stable maximum frequency ~1.0–1.1 seconds per measurement; suitable for low-duty-cycle battery-powered operation.
- Stable operation requires ≥ 50 ms delay for LTE-M radio module and Modbus polling.
- JSON assembly time must be ≥ 100 ms for reliable posting.
- All success criteria met, confirming that the LTE-M implementation is production-ready for Phase 2 industrial-grade design.

Lessons Learned

- Optimal firmware delay values are critical for LTE-M stability.
- JSON assembly time significantly impacts posting reliability.
- VxLAN tunneling for LoRaWAN integration is transparent to devices but adds robust security benefits.
- 5G lab coordination accelerated troubleshooting and validation.

Comparisons and Industry Context

This initiative embodies ADNOC's commitment to UAE In-Country Value (ICV) by fostering local R&D and manufacturing capabilities. The hardware, firmware, and system integration were executed domestically by GOEasyFlow, with multi-vendor collaboration from Etisalat, Nokia, and Siemens. Local manufacturing not only reduces supply chain lead times but also strengthens the UAE's industrial base in advanced IoT solutions.

Table 3—Qualitative Cost, Commissioning, and Cybersecurity Comparison

Technology	Cost (Qualitative)	Commissioning Time	Cybersecurity Posture
LoRaWAN	Low	Weeks	Medium – requires additional security layers
LTE-M	Low	Weeks	High – benefits from 5G core security
Wired	High (2–10× wireless)	Months–Years	High – physically isolated

Table 4—Battery Life Expectations*

Technology	Typical Battery Capacity (Messages)	Typical Battery Capacity (Years)
LoRaWAN	~120,000 messages	More than 5 years @30 min frequency, >2 years @10 min, >1 years @5 min
NB-IoT	~105,000 messages	~10% less than LoRaWAN under same conditions
LTE-M	~85,000 messages	~30% less than LoRaWAN under same conditions

* Estimates assume normal field conditions, one 16,500 mAh Lithium battery. Capacity can be duplicated by using two batteries instead of one.

Project Phases and Milestones

Phase 1 – Achievements

- Q1 2025: ADNOC & GOEasyFlow started to develop 5G instrumentation suitable for monitoring and control activities in upstream operations.
- Q2 2025: Development of the GOEasyFlow prototype for lab testing of a 5G-enabled device for monitoring and control.
- July 2025: Testing at ADNOC's 5G Lab of the prototype using LoRaWAN backboned with a 5G VxLAN tunnel (Nokia FastMile mm29-b).
- August 2025: Testing at ADNOC's 5G Lab of the prototype using LTE-CatM Low Band (700 MHz), demonstrating stable operation and meeting all success criteria.

Planned Future Phases

The next phases will progressively mature the technology from lab prototype to fully certified field instrumentation:

- Phase 2 (Q1 2026): Design, manufacturing, and laboratory testing of an industrial-grade 5G IoT battery-powered instrument for use in non-ATEX environments.
- Phase 3 (Q3 2026): ATEX certification of the new instrumentation to allow deployment in hazardous areas.
- Phase 4 (Q4 2026): Field rollout of the certified instrumentation across ADNOC's operational sites.

Acknowledgments

The authors thank ADNOC D&CS group, ADNOC Upstream, ADNOC's 5G Lab, Accenture, Etisalat, Nokia, and Siemens for their collaboration, and the GOEasyFlow engineering team for device development and testing.

Nomenclature

LTE-M (Cat-M1)	A 3GPP-standardized low-power wide-area (LPWA) technology designed for IoT.
LoRaWAN-over-5G	Encapsulation of LoRaWAN traffic within a 5G network tunnel, allowing existing LoRaWAN devices to benefit from 5G security.
VxLAN	
Edge Computing	Performing control and processing logic locally within field devices to minimize latency.
Modbus RTU	A widely used serial communication protocol in industrial environments.
MQTT	Lightweight publish-subscribe messaging protocol optimized for constrained devices and low-bandwidth networks.
CPE	Customer Premises Equipment (on-site device like a modem/router)
RRU	Remote Radio Unit (part of the base station handling RF signals)
BTS	Base Transceiver Station (cellular radio equipment)
5GPN	5G Private Network
GNb	gNodeB (5G base station)
RLE	Radio Link Enabler (Nokia access aggregator in this setup)
CLE	Core Link Enabler (Nokia element connecting access to core)
CMU	Core Management Unit (part of Nokia's 5G core platform)
OT FW	Operational Technology Firewall (segregates industrial network from IT network)
CDS	Central Data System (control room/server side)
IPSec	Internet Protocol Security (secure tunneling encryption)
GTP	GPRS Tunneling Protocol (used in mobile cores for user data transport)
L2 SW	Layer 2 Switch (Ethernet switch operating at Data Link layer)
POE	Power Over Ethernet (power and data over same Ethernet cable)